

User Guide of Chandrayaan-2 DFSAR Level-3C Polar Mosaic Products



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Document Control & Data Sheet

1.	Title :	User Guide of Chandrayaan-2 DFSAR Level-3C Polar Mosaic Products
3.	Type of Report :	Technical
5.	No. of Pages:	10
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7.	Originating Unit :	Application Techniques Development Division, Application Techniques Development and Validation Group
8.	Abstract :	This document provides the strategy to generate the Level-3C Polar Mosaic products, along with the different type of products and description of nomenclature and archival strategy.
9.	Keywords :	Chandrayaan-2, PDS4, Deriver, Polar Mosaic
10.	Security Classification :	Unrestricted
11.	Distribution:	Users of DFSAR Data

1.1 DSFAR Data levels:

Chandrayaan-2 Dual Frequency SAR is the world's first dual frequency (L-band and S-band) moon mission launched on 22nd July 2019. Dual-frequency SAR on-board Chandrayaan-2 aims to address a host of applications to understand the scattering characteristics of lunar surface and subsurface features; to map lunar craters and other features especially in the polar regions with finer details; to locate and quantify water-ice in the lunar polar regions and retrieve lunar surface parameters like dielectric constant and surface roughness. Chandrayaan-2 Dual Frequency SAR operates in StripMap mode with a nominal swath of 10 Km. Data acquired in various modes [single/hybrid(CP)/full-polarimetric(FP)] are available for use. Majorly the data acquired in full-polarimetric (FP) mode, in which transmit polarization is switched between V & H for consecutive pulses and reception is by simultaneous dual-linearly polarized channels, for each of the transmit polarization resulting to HH, HV, VH and VV radar backscatter coefficients. Data processing is done based on the defined levels. There are four level defined for Chandrayaan-2 SAR data products i.e. Level-0, Level-1, Level-2 and Level-3, which are described in Table 1.

Table 3.2 Chandrayaan-2 PDS4 Data Processing Levels

PDS4 Level	PDS4 Level Description	ISRO Level	CODMAC Level
Raw	Original data from the SAR instrument. If compression (BAQ), reformatting, packetization, or other translation has been applied to facilitate data transmission or storage, those processes will be reversed so that the archived data are in a PDS approved archive format. Level0A comprises of raw signal data and Level-0B contains onboard range compressed data.	0A,0B	2
Calibrated	Product will have information for conversion from data to physical units, which makes values independent of the instrument. Level-1A consist of Seleno-tagged SLC slant range product (sli). Level-1B consist of Seleno-tagged ground range (gri) product. Level-2 consist of Seleno-Referenced (sri) product.	1A,1B,2	3

Derived	Level-3A CP: 3 Layers (2 real Diagonal: 1 complex Off Diagonal Elements of COVariance Matrix) FP: 6 Layers (3 Real Diagonal: 3 Complex Off Diagonal Elements of COVariance Matrix) Level-3B Seleno-Referenced Layers of Level-3A data product Level-3C 1. Multiple Level-3B product layers Mosaiced into a North Pole or South Pole Mosaic product for an Imaging mode. 2. Level-3B Covariance matrix products are utilized to generate various radar polarimetric parameters. The Level-2 FP radar backscatter values are utilized to generate calibrated backscatter coefficients and inversion parameters.	3A,3B,3C	4+
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1.2 Description of DFSAR Derived polarimetric parameters:

With the availability of full-polarimetry data from Chandrayaan-2 Dual-frequency SAR (DFSAR), the lunar surface can be studied in a new perspective using unique set of radar derived polarimetric child parameters. Along with traditional Circular Polarization Ratio (CPR), which is sensitive to both surface roughness and presence of water ice, Single-bounce Eigen value Relative Difference (SERD) and a ratio based on diagonal elements of coherency matrix (T3) termed as T-Ratio are the two parameters are also derived. SERD has monotonically varying response thereby allowing for unambiguous descriptor of surface roughness, and hence applied for any DFSAR analysis. A low value of SERD would indicate a rough surface, whereas a high value would correspond to a smooth surface. On the other hand, T-ratio is independent on the surface roughness and depends only on the dielectric constant of the surface and the incidence angle. Increase in T-Ratio values indicate increase in Dielectric Constant.

The idea of the polarimetric SAR decomposition is to understand the naturally distributed targets using physical scattering mechanisms (odd, even, volume and helix

scattering). For instance, the different cases of: moderately smooth surface (where the surface roughness is small compared to the radar wavelength), a pair of orthogonal surfaces (dihedral structure) with different dielectric constants and inhomogeneous bulk material that contains local variations in dielectric properties can give rise to odd, even and volume scattering, respectively. Besides, a helix target or two or more diplanes can generate a circularly polarized scattered beam from a linearly polarized incident beam leading to helix scattering. The polarimetric decomposition elements are derived as it is helpful to understand the radar scattering behaviour from the lunar surface features.

1.3 Generation strategy of Level-3C Polar Mosaic products:

Individual Level-3B Seleno-referenced Map-Projected Covariance matrix products of L-band full-polarimetric (FP) DFSAR data are utilized to generate various radar polarimetric parameters. Similarly, the Level-2 Seleno-referenced full-polarimetry radar backscatter values in different polarization channels (HH, HV, VH and VV) are utilized to generate calibrated backscatter coefficients and inversion parameters. Thus, the generated polarimetric parameters, calibrated backscatter coefficients and inversion parameters are listed below:

- a. Circular Polarization Ratio (CPR)
- b. Single bounce Eigenvalue Relative Difference (SERD)
- c. T-Ratio
- d. Even bounce scattering backscatter
- e. Odd bounce scattering backscatter
- f. Volume scattering backscatter
- g. Helix scattering backscatter
- h. Calibrated HH-pol backscatter coefficient (σ_{HH}^0)
- i. Calibrated HV-pol backscatter coefficient (σ_{HV}^0)
- j. Calibrated VH-pol backscatter coefficient (σ_{VH}^0)
- k. Calibrated VV-pol backscatter coefficient (σ_{VV}^0)
- l. Dielectric Constant
- m. Root Mean Square Height
- n. Correlation Length

The algorithms utilized to generate Circular Polarization Ratio (CPR), Single bounce Eigenvalue Relative Difference (SERD) and T-Ratio parameters from full-polarimetric DFSAR data are described in Putrevu et al., (2023). Similarly, for generation of polarimetric decomposition products, Yamaguchi 4-component scattering power decomposition with rotation of coherency matrix (Y4R) method (Yamaguchi et al., 2011) is utilized. The calibrated backscatter coefficients are generated from backscatter values of different polarization channels using the following equation:

$$\sigma^{\circ}[\text{dB}] = 20\log_{10}(\text{DN}) - \text{Calibration Constant}$$

The Dielectric Constant, Root Mean Square Height and Correlation Length values are estimated using a physical inversion model (Pandey et al., 2023) based on Advanced Integral Equation Model (Chen et al., 2003).

These parameter layers are mosaicked into a North pole or South pole mosaic product for a particular Imaging look direction as Level-3C data product. DFSAR datasets are available in left and right looking mode in ascending and descending pass. To maintain the homogeneity with the mosaic products available from previous missions (Mini-SAR in Chandrayaan-1, Mini-RF in Lunar Reconnaissance Orbiter), the polar mosaics are generated in both East looking and West looking mode. Hence, for East and West looking mode the following imaging configurations are utilized:

East looking	West looking
Right look in ascending pass	Left look in ascending pass
Left look in descending pass	Right look in ascending pass

Datasets with undesirable artefacts due to dynamic update and radiometric issues are not considered for mosaicking. Besides, the datasets which are acquired in nominal incidence angle (look angle) of 26°, are utilized for mosaicking to maintain the uniformity. Thus, mosaic products corresponding to different radiometric, polarimetric and inversion parameters listed previously, are generated for both north and south polar regions in both east looking and west looking mode. Similar to Level-3B, these Level-3C polar mosaic products are in Polar-Stereographic map projection at 25-meter pixel spacing.

1.4 Archival strategy of the Level-3C products:

The **Level-3C mosaic products** are archived majorly in **packaging folders**, which are named in the following manner.

- ❖ **Polarimetric Child Product (pcp):** This packaging folder contains mosaic of three polarimetric child products (pcp) namely CPR (cpr), SERD (srd) and T-Ratio (trt) for individual pole in separate look direction. The mosaic products are in GeoTiff format and the corresponding PDS4 complied common metadata file of these three mosaic products is provided in xml format (Figure 1).

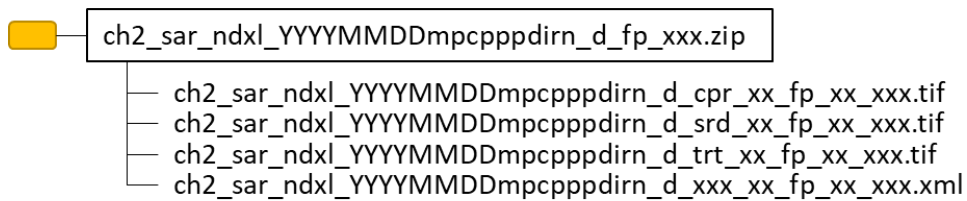


Figure 1: The mosaic products present in polarimetric child product (pcp) folder.

- ❖ **Polarimetric Decomposition Product (y4r):** This packaging folder contains mosaic of four polarimetric decomposition products using Y4R method (y4r) namely Even (evn), Volume (vol), Odd (odd) and Helix (hlx) for individual pole in separate look direction. The mosaic products are in GeoTiff format and the corresponding PDS4 complied common metadata file of these four mosaic products is provided in xml format (Figure 2).



Figure 2: The mosaic products present in polarimetric decomposition (y4r) folder.

- ❖ **Calibrated Backscatter Coefficient Product (cbc):** This packaging folder contains mosaic of four calibrated radar backscatter coefficients (cbc) corresponding to HH-polarization (hh), HV-polarization (hv), VH-polarization (vh) and VV-polarization (vv) for individual pole in separate look direction. The mosaic products are in GeoTiff

format and the corresponding PDS4 complied common metadata file of these four mosaic products is provided in xml format (Figure 3).

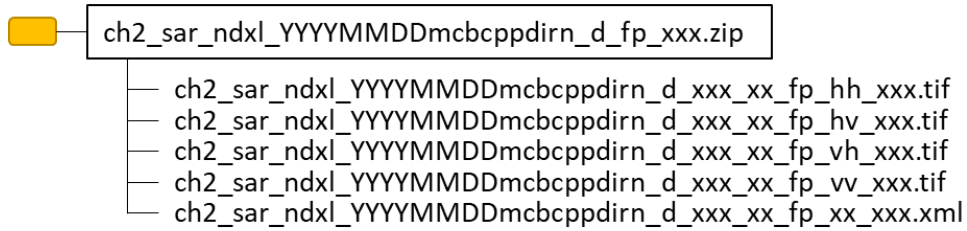


Figure 3: The mosaic products present in radar backscatter product (cbc) folder.

- ❖ **Inversion Product (inv):** This packaging folder contains mosaic of three inversion products (inv) namely Dielectric Constant (rdp), Root Mean Square Height (rms) and Correlation Length (lcr) for individual pole in separate look direction. The mosaic products are in GeoTiff format and the corresponding PDS4 complied common metadata file of these three mosaic products is provided in xml format (Figure 4).



Figure 4: The mosaic products present in inversion product (inv) folder.

The acronyms utilized to name the packaging folders and the individual mosaic products are described below:

<ch2>	Chandrayaan-2
<sar>	DFSAR
<d>	derived
<l>	L-band
<YYYYMMDD>	Year Month Date of product generation
<m>	Mosaic product
<pcp>	Polarimetric child product
<y4r>	Polarimetric Decomposition product (Yamaguchi Decomposition-Y4R)

<cbc>	Calibrated backscatter coefficient product
<inv>	Inversion product
<pp>	Pole ('np' or 'sp' representing north pole or south pole respectively)
<dirn>	Look direction ('east' or 'west')
<d>	data
<cpr>	CPR
<srd>	SERD
<trt>	T-Ratio
<evn>	Even
<vol>	Volume
<odd>	Odd
<hlx>	Helix
<fp>	Full-polarimetric
<hh>	Calibrated HH-pol backscatter coefficient
<hv>	Calibrated HV-pol backscatter coefficient
<vh>	Calibrated VH-pol backscatter coefficient
<vv>	Calibrated VV-pol backscatter coefficient
<rdp>	Dielectric Constant
<rms>	Root Mean Square Height
<lcr>	Correlation Length

Apart from the GeoTiff mosaic product files and corresponding metadata (xml) file, the packaging folders contain browse images (png). In the 'pcp', 'y4r', 'cbc' and 'inv' packaging folders the browse images are represented by mosaics of CPR, Volume, Calibrated HH-backscatter coefficient and Dielectric constant, respectively. The list of Level-1 datasets utilized to generate the Level-3C mosaic products are listed in the 'foldername.txt' file in the 'miscellaneous' sub-folder. The structure of the packaging folders is displayed in Figure 3.

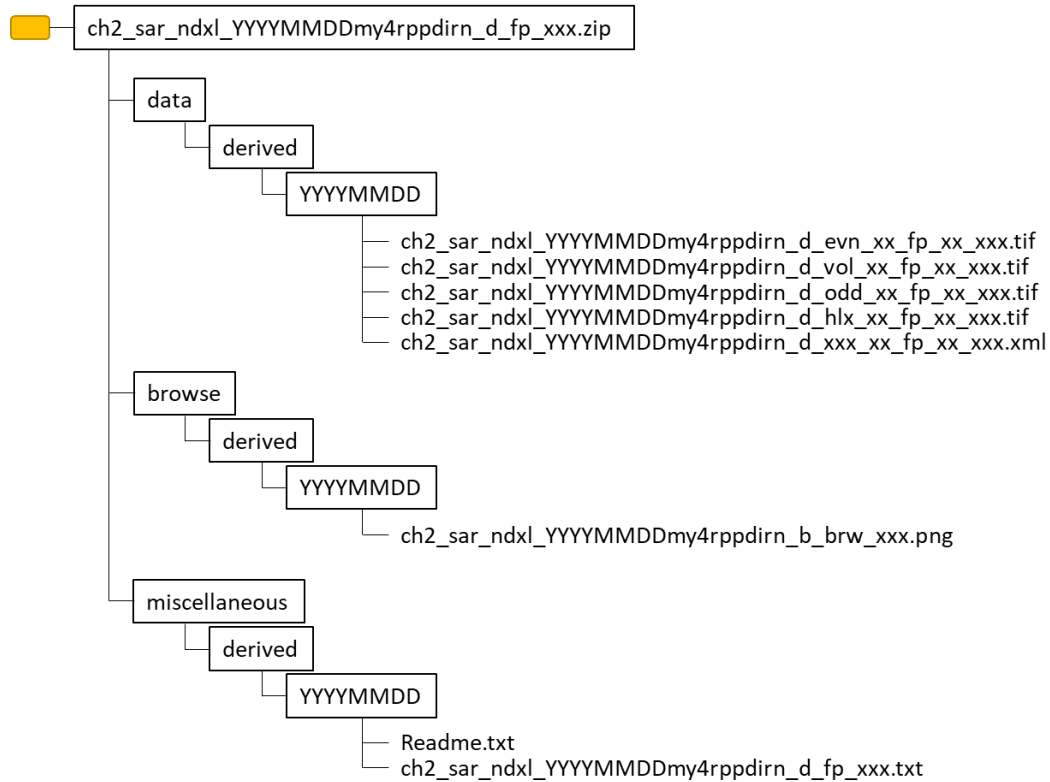


Figure 3: Structure of the packaging folders and the files present within the sub-folders.

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